

Demonstration of Proposed Features

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Data Engineering	Processes soil and environmental data (N, P, K, temperature, humidity, pH, rainfall) and prepares it for model training.	Implemented	Yes	Cleaned dataset, handled missing values, encoded labels, and structured features for ML pipeline.
2	Exploratory Data Analysis (EDA)	Generates visual insights such as distribution of nutrients, rainfall patterns, and crop-wise analysis.	Implemented	Yes	Used Matplotlib and Seaborn to generate histograms, countplots, and correlation analysis graphs.
3	Data Preprocessing Pipeline	Applies scaling, transformation, and train test split for model preparation.	Implemented	Yes	Used Pandas + Scikit-learn preprocessing and train_test_split for structured ML pipeline.
4	Model Training & Selection	Trains multiple ML models like Logistic Regression, KNN, Decision Tree, Random Forest, and K-Means clustering.	Implemented	Yes	Random Forest and Logistic Regression used for final prediction; K-Means used for grouping soil patterns.
5	Web Application (Flask Interface)	Provides a user-friendly interface to input soil parameters and get crop recommendations.	Implemented	Yes	Fully functional Flask app with HTML forms and prediction output display.
6	Crop Prediction System	Predicts the most suitable crop based on environmental conditions using trained ML model.	Implemented	Yes	Real-time prediction achieved using saved model.pkl integrated with Flask backend.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100